

Technical Indicators & Strategy Backtesting

Ticker: AAPL | 5 years of daily data

FINM 25000 — Homework 2

Strategies

STRATEGY DESCRIPTIONS & RULES

Strategy 1 – Trend Following (MACD + ADX)

Idea: ride established trends; only act when a real trend is present.

Buy / hold long when: MACD > Signal line AND ADX > 25

Exit to flat when: MACD < Signal line

ADX acts as a filter so MACD crossovers in choppy, trendless markets are ignored.

Strategy 2 – Mean Reversion (RSI + Bollinger Bands)

Idea: fade extremes; buy when price is stretched down, sell when stretched up.

Buy when: RSI < 30 AND Close < lower Bollinger Band

Sell when: RSI > 70 OR Close > upper Bollinger Band

Strategy 3 – Custom "Confirmed Trend" (EMA + RSI + OBV)

Combines three indicator categories: trend, momentum, and volume.

Buy when: EMA(20) > EMA(50) AND RSI < 70 AND OBV > OBV's 20-day SMA

Sell when: EMA(20) < EMA(50) OR RSI > 80

Only takes the trend trade when it is not already overheated (RSI) and when volume flow confirms buying pressure (OBV).

Benchmark – Buy & Hold

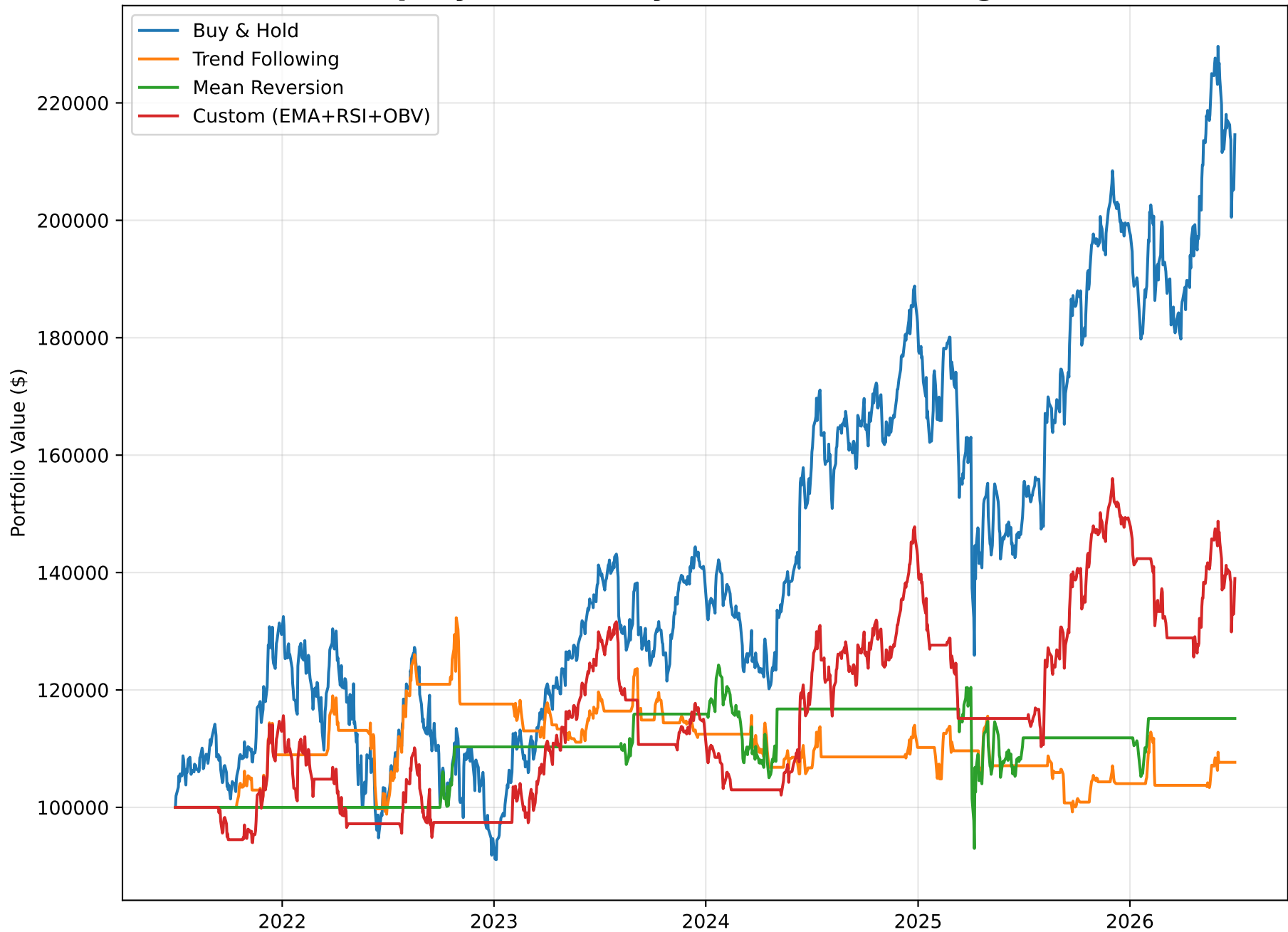
Buy on day one, hold the whole period. The bar every active strategy must beat on a risk-adjusted basis.

Backtest assumptions: \$100,000 initial capital, long-only, no leverage, no shorting. Signals are traded on the next day's close to avoid lookahead bias.

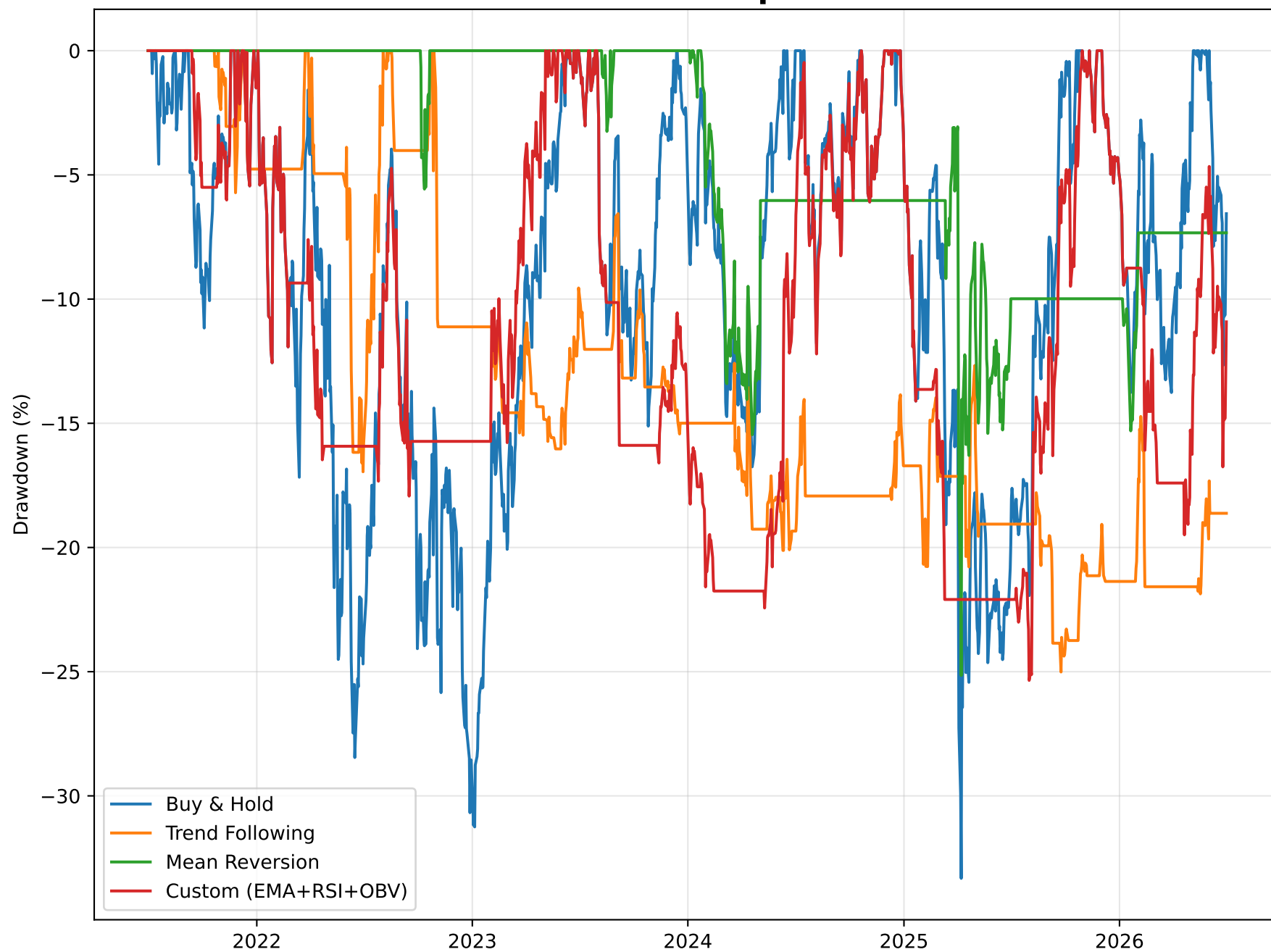
Performance Comparison

	Total Return	CAGR	Volatility (ann.)	Sharpe Ratio	Sortino Ratio	Max Drawdown	Win Rate	# Trades
Buy & Hold	114.58%	16.50%	27.59%	0.62	0.90	-33.32%	—	—
Trend Following	7.67%	1.49%	13.74%	0.03	0.03	-25.01%	45.71%	35
Mean Reversion	15.14%	2.86%	13.85%	0.13	0.08	-25.15%	100.00%	5
Custom (EMA+RSI+OBV)	38.99%	6.81%	18.24%	0.34	0.39	-25.35%	46.67%	15

Equity Curve: Buy & Hold vs. Strategies



Drawdown Comparison



AAPL — Price & Signals (Trend Following)



Discussion

DISCUSSION OF RESULTS

On a risk-adjusted basis, Buy & Hold had the highest Sharpe ratio (0.62).
The highest total return came from Buy & Hold (114.58%).
Trend Following experienced the smallest maximum drawdown (-25.01%).

Which strategy performs best depends on what you optimize for:
raw return, risk-adjusted return (Sharpe/Sortino), or capital
preservation (drawdown). A strategy can beat Buy & Hold on Sharpe
while trailing it on total return if it sidesteps the worst
drawdowns by moving to cash.